

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE CODE: ITA 0606

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO1: *Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)*

Assessment Tool Weightage: 10%

QUESTIONS: 15 Total Marks: 25 Annexure A

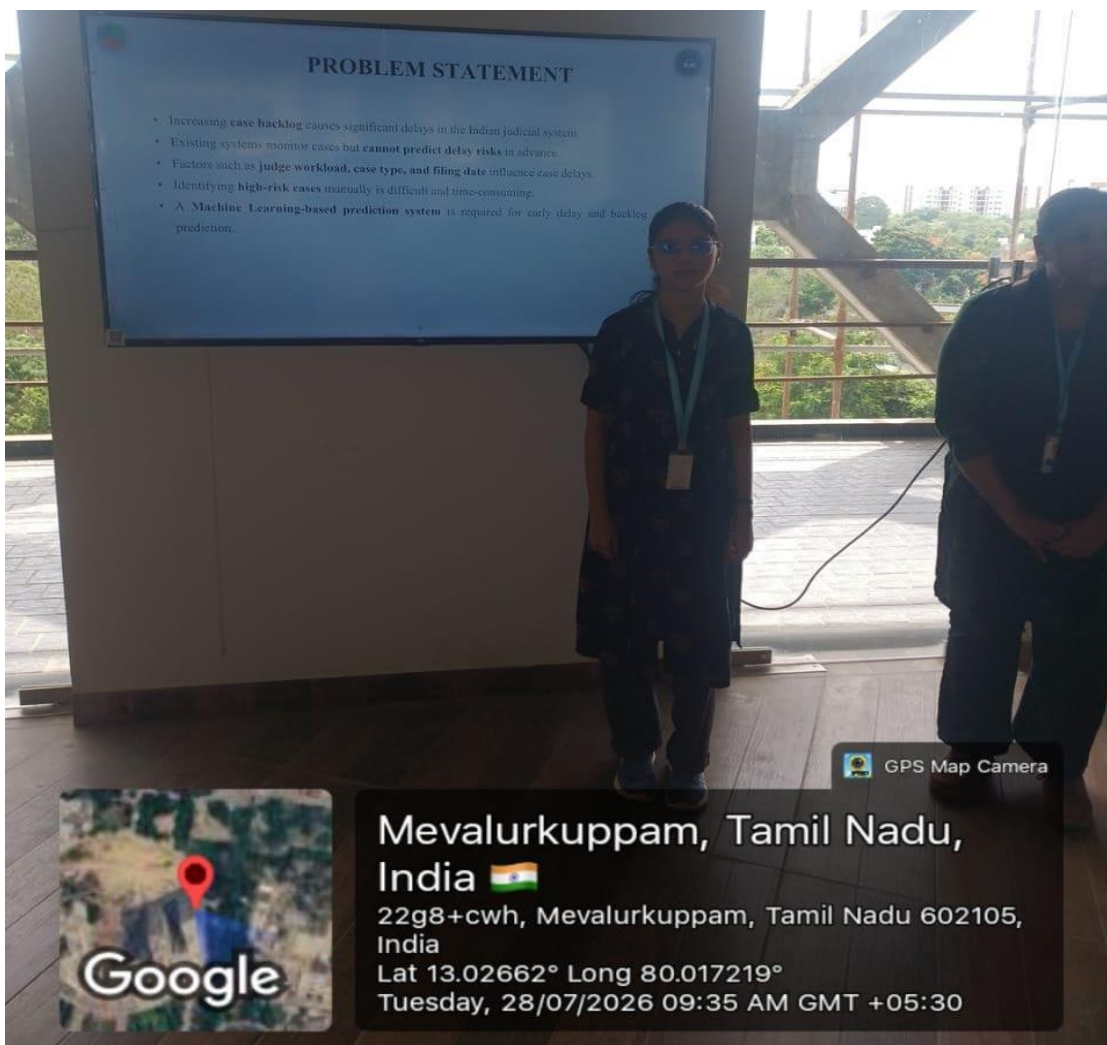
TECHNICAL PRESENTATION

Code of Conduct:

I _____ Chalamala Sai Harshitha (192424346)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: C. Sai Harshitha





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Machine Learning-Based Early Prediction of Case Backlog and Delay Risk in the Indian Judicial System

ITA0606-MACHINE LEARNING

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Abstract



- Develops a **Machine Learning** system for early prediction of judicial case delays.
- Processes historical court data using **preprocessing** and **feature engineering**.
- Applies **Logistic Regression, Random Forest, and XGBoost** models.
- Predicts **case delay risk, court backlog, and delay duration**.
- Uses **SHAP** to identify key factors influencing delays.
- Deploys the model as a **Streamlit web application** for real-time predictions.

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Problem Statement



- Increasing **case backlog** causes significant delays in the Indian judicial system.
- Existing systems monitor cases but **cannot predict delay risks** in advance.
- Factors such as **Judge workload, case type, and filing date** influence case delays.
- Identifying **high-risk cases** manually is difficult and time-consuming.
- A **Machine Learning-based prediction system** is required for early delay and backlog prediction.

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Objectives



- To develop an ML model for predicting case delay risk and court backlog.
- To analyze judicial data to identify delay patterns.
- To compare Logistic Regression, Random Forest, and XGBoost models.
- To identify high-risk cases using feature importance.
- To evaluate model performance using standard metrics.
- To deploy the model as a Streamlit web application.

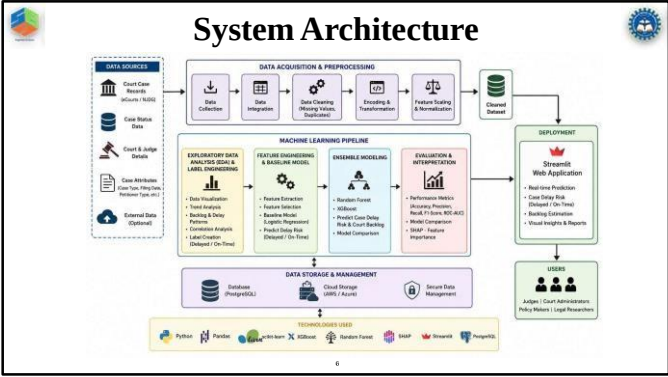
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Literature Survey

Sl. No	Year	Authors	Title of the Publication	Findings	Limitations
1	2021	Kshitiz Verma et al.	Estimating Time to Clear Pendency of Cases in High Courts in India	Predicted court backlog using NIDG data.	No case-level prediction.
2	2023	Mohit Bhatnagar et al.	Predicting Delays in Indian Lower Courts Using AutoML	ML improved delay prediction.	Limited to lower courts.
3	2024	Trivedi Vivek et al.	AI in the Indian Judiciary	AI can help reduce case backlog.	No predictive model developed.
4	2025	Rahul K. Bharati	Predictive Justice in Indian Courts	ML enhanced judicial predictions.	Focused on case outcomes.
5	2026	Mohit Bhatnagar et al.	Hybrid ML for Predicting Case Delays	XAI improved prediction accuracy.	Requires large datasets.

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Module 1

Data Acquisition & Preprocessing

- Collect historical judicial case data from court records.
- Gather attributes such as:
 - Case Type, Court, Filing Date, Backlog, and Judge Workload.**
- Handle missing values and remove duplicate records.
- Encode categorical attributes and normalize numerical data.
- Generate a clean dataset for EDA and machine learning.

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Module 2

Exploratory Data Analysis (EDA) & Label Engineering

- Analyze relationships among:
 - Case Type, Court, Filing Date, Backlog, and Judge Workload.**
- Visualize case trends, backlog distribution, and delay patterns.
- Identify key factors influencing judicial delays.
- Create target labels as **Delayed** or **On-Time**.
- Generate insights for feature engineering and model training.

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Module 3



Feature Engineering & Baseline Model

- Extract meaningful features from the processed judicial data.
- Select key attributes for model training.
- Train a **Logistic Regression** baseline model.
- Predict **case delay risk** as **Delayed** or **On-Time**.
- Evaluate baseline performance for further improvement.

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Module 4



Ensemble Modeling

- Train **Random Forest** and **XGBoost** ensemble models.
- Learn complex patterns from judicial case data.
- Predict **case delay risk** and **court backlog**.
- Compare model performance with the baseline model.
- Select the best-performing model for deployment.

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Module 5



Evaluation & Deployment

- Evaluate the model using **Accuracy, Precision, Recall, F1-Score, and ROC-AUC**.
- Compare the performance of all trained models.
- Interpret predictions using **SHAP feature importance**.
- Deploy the best-performing model using **Streamlit**.
- Enable real-time prediction of **case delay risk** and **court backlog**.

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Conclusion



- Developed an ML-based system to predict **case delay risk** and **court backlog**.
- Improved prediction accuracy using **Random Forest** and **XGBoost** models.
- Identified key factors influencing judicial delays through **SHAP** analysis.
- Evaluated the model using standard performance metrics.
- Deployed the solution as a **Streamlit** web application for real-time predictions.
- Supports efficient judicial planning and faster decision-making.

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References

1. Verma, K., Musaddi, A., Mittal, A., & Jain, A. (2024). *Estimating time to clear pendency of cases in High Courts in India using linear regression on NJDG data*. *Journal of Open Access to Law*.
2. Bhatnagar, M., & Huchhanavar, S. S. (2024). *Predicting delays in Indian lower courts using AutoML and decision forests*. In *Computing, Internet of Things and Data Analytics* (Studies in Computational Intelligence).
3. Trivedi, V., & Nilakshi. (2024). *Artificial intelligence in the Indian judiciary: A systematic analysis of potential applications and challenges in addressing case backlogs*.
4. Bharati, R. K. (2025). *Predictive justice in Indian courts: Machine learning approaches to case outcome forecasting*. *SSRN Electronic Journal*.
5. Bhatnagar, M., & Huchhanavar, S. S. (2026). *Hybrid machine learning modelling with explainability for predicting case delays and durations in Indian lower courts*. *Journal of Big Data*, 13.



Thank you